Department of Computer Science & Information Technology

CT-261 Database Management System

Complex Computing Activity (CCA) through Open-Ended Lab (OEL)

Course Learning Outcome:

(CLO -3): Implement basic programming solutions to simple real-world problems using appropriate logic and programming constructs. (C3, PLO-4)

Complex Computing Activity Attributes:

1. Level of Interactions (CA-2): The Database Management System resolves complex interactions among technical, computing, and contextual components such as sensors, utility services, maintenance teams, and citizen requests by enforcing business rules, maintaining data integrity, and supporting analytical decision-making.
2. Innovation (CA-3): The system demonstrates innovation through the creative application of database and domain principles by integrating sensor data, resource management, and incident analytics into a unified and intelligent smart city management solution.
3. Familiarity (CA-5): This project extends beyond familiar problems by applying fundamental DBMS concepts such as normalization and integrity constraints to a new and complex real-world smart city environment involving multiple stakeholders and dynamic data sources.

Problem Statement

Apply database design and development concepts to develop a database management system (DBMS) solution.

Description

You will select a project, analyze the organizational issues and needs, and work your way through the conceptual, logical, and physical designs of a DBMS solution. This will require substantial research of best practices in design, available products, and the legal and ethical standards to which you must adhere during design. The skills required in this assessment will be valuable in the role of an IT manager or DBMS professional, as these individuals are often tasked with developing solutions to various organization data problems while also adhering to legal, ethical, and financial considerations. The ability to craft and present professional and logical proposals will also be expected of IT professionals, especially those at the management level.

The following critical elements must be addressed:

I. **Organization Problem:** Analyze the organization to determine the problem/challenge, business requirements, and limitations of current system(s).

II. **Analysis and Design**
   A. **Conceptual Model:** Based on the business problem or challenge, devise a conceptual model that would best address the problem. Your model should include all necessary entities, relationships, attributes, and business rules.
   B. **Logical Model:** Based on the conceptual model, illustrate a logical model for your DBMS that accurately represents all necessary aspects of the DBMS to address the solution.
   C. **Physical Design:** Create a physical database design that builds on the nonphysical (conceptual and logical) models you crafted.

Deliverables

- **Code file : contain the code of database creation.**
- Prepare a report and bring it at the day of your viva. The report has to be organized as follows:
  Title page. The rubric sheet given at the end of this document will serve as the title page of your report.
  Text of 300 words at maximum, containing the following:

1. Business requirements, problem/challenge, and limitations of current system
2. Participation in a team
3. ER and EER Diagram of your project.
4. Functional Dependencies
5. Process of Normalization (1NF to BCNF)
6. Sample Queries

Suggested Domain Ideas (Examples Only):

- Innovation & Startup Incubation Ecosystem
- Smart City Infrastructure & Resource Monitoring
- E-Sports Tournament & Performance Analytics Platform
- Digital Content Creator Monetization System
- Environmental Compliance & Sustainability Tracking System
- Global Freelance & Skill Marketplace Platform (Other innovative domains may also be proposed.)

Assumptions & Constraints:

- You are required to complete the task in your designated groups.
- A student should not be involved in more than 2 active projects simultaneously.
- The system must include at least 8–10 core entities.
- At least two different field types from the following list must be included: numeric, alpha, date/time, and currency.
- A minimum of three many-to-many relationships must be modeled.
- At least five meaningful business rules must be defined and enforced.
- Appropriate primary key, foreign key, and integrity constraints must be implemented.

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Student Name: ____________________ Student Roll No.: ____________________

Criteria	0 – Bad (0 Marks)	1 – Average (1 Mark)	2 – Good (2 Marks)
1. Domain Understanding & Problem Definition	Domain is unclear, trivial, or poorly defined with no real-world relevance.	Domain is partially defined with limited complexity and weak problem description.	Domain is clearly defined, complex, and represents a realistic real-world system with clear objectives.
2. Conceptual Design (ER Diagram)	ER diagram is missing, incorrect, or shows very few entities and relationships.	ER diagram includes basic entities and relationships but lacks clarity or completeness.	ER diagram clearly represents all required entities, relationships, and cardinalities accurately.
3. Normalization & Logical Design (3NF)	Tables are not normalized and show redundancy and anomalies.	Partial normalization is applied but some dependencies or anomalies remain.	Database schema is fully normalized up to 3NF with no insertion, deletion, or update anomalies.
4. Business Rules & Integrity Constraints	Business rules are missing or not enforced using constraints.	Some business rules are defined but weakly enforced.	Multiple meaningful business rules are clearly defined and enforced using primary keys, foreign keys, and constraints.
5. Analytical Queries & Innovation	Queries are missing or do not support decision-making; design shows no innovation.	Basic queries are provided with limited analytical value.	Meaningful analytical queries are included and the design shows innovative and creative application of DBMS principles.

Total Marks: ______/10____

Teacher's Signature: ____________